Chapter Four: Discrete Fourier Transform (DFT)

Content

- Frequency Domain Sampling
- Relationship of the DFT to the Fourier transform
- Discrete Fourier Transform
- Properties of the DFT

Discrete Fourier Transform

- Frequency analysis of discrete-time signals must conveniently be performed on a computer or DSP.
- Recall:

$$\text{aperiodic in time} \overset{F}{\leftrightarrow} \text{continuous in frequency}$$

$$x(n) \overset{F}{\leftrightarrow} X(\omega)$$

- $X(\omega)$ cannot therefore be computed for the entire set $\omega \in \mathbb{R}$; for practicality, $X(\omega)$ must be computed for a discrete and finite set of values in $\omega \in \mathbb{R}$.

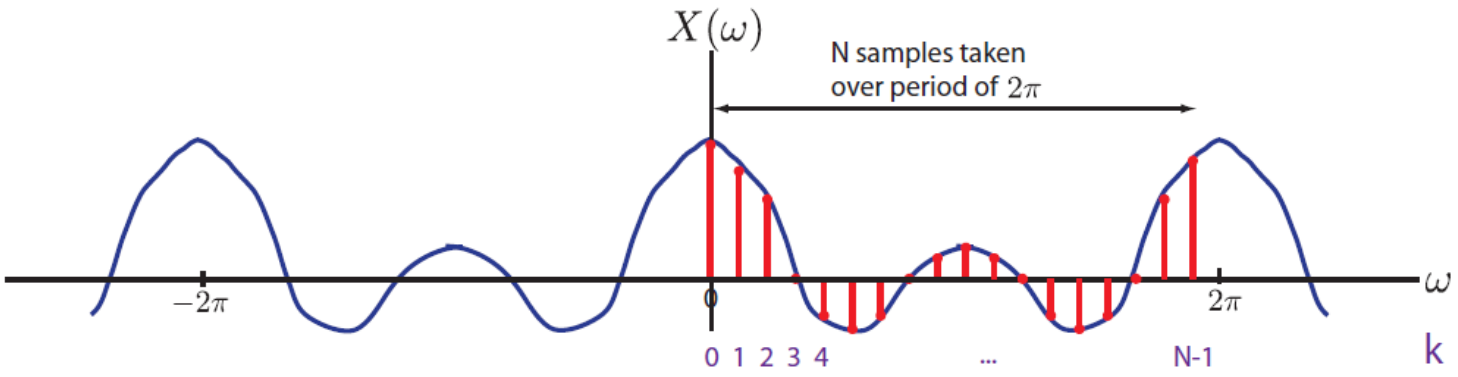
Discrete Fourier Transform

- Strategy to compute $X(\omega)$:
  1. Compute $X(\omega)$ for equally spaced samples.
  2. Compute $X(\omega)$ samples for one period only (recall, $X(\omega)$ is periodic with period 2π)
- Assuming we compute N samples of $X(\omega)$ over one period of 2π , the resulting computed frequency signal would respectively be a sampled version of $X(\omega)$ such that:

$$\omega = \frac{2\pi k}{N}$$

Frequency Domain Sampling

$$\omega = \frac{2\pi k}{N}$$



- Note: spacing between each sample is $\frac{2\pi}{N}$.

Frequency Domain Sampling

- Recall, the discrete-time Fourier transform (DTFT):

$$x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\omega)e^{j\omega n}d\omega$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$$

- Suppose we sample $X(\omega)$ according to: $\omega = \frac{2\pi k}{N}$

Frequency Domain Sampling

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$X\left(\frac{2\pi}{N}k\right) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\frac{2\pi}{N}kn},$$

$$= \sum_{l=-\infty}^{\infty} \sum_{n=lN}^{lN+N-1} x(n) e^{-j2\pi k\frac{n}{N}} \quad \text{Let } n' = n - lN$$

$$= \sum_{l=-\infty}^{\infty} \sum_{n'=0}^{N-1} x(n'+lN) \underbrace{e^{-j2\pi k\frac{n'+lN}{N}}}_{=e^{-j2\pi k\frac{n'}{N}}\underbrace{e^{-j2\pi k\frac{lN}{N}}}_{=1}}$$

Frequency Domain Sampling

- For $k = 0, 1, 2, \ldots, N-1$,

$$X\left(\frac{2\pi}{N}k\right) = \sum_{l=-\infty}^{\infty}\sum_{n=0}^{N-1} x(n+lN)e^{-j2\pi k\frac{n}{N}}$$

$$= \sum_{n=0}^{N-1}\sum_{l=-\infty}^{\infty} x(n+lN)e^{-j2\pi k\frac{n}{N}}$$

$$= \sum_{n=0}^{N-1} \underbrace{\left[\sum_{l=-\infty}^{\infty} x(n+lN)\right]}_{\text{equivalent signal } x_p(n)} e^{-j2\pi k\frac{n}{N}}$$

$$= \sum_{n=0}^{N-1} x_p(n)e^{-j2\pi k\frac{n}{N}}$$

where $x_p(n)$ is the periodic repetition of $x(n)$.

Frequency Domain Sampling

For $k = 0, 1, 2, \ldots, N-1$,

$$X\left(\frac{2\pi}{N}k\right) = \sum_{n=0}^{N-1} x_p(n) e^{-j2\pi k \frac{n}{N}}$$

- Characteristics of $x_p(n)$: (1) discrete-time, (2) period $= N$, (3) has as a DTFS:

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x_p(n) e^{-j2\pi k \frac{n}{N}}$$

- $X\left(\frac{2\pi}{N}k\right)$ looks like a scaled DTFS of $x_p(n)$,

$$c_k = \frac{1}{N} X\left(\frac{2\pi}{N}k\right) \quad k = 0, 1, \ldots, N-1.$$

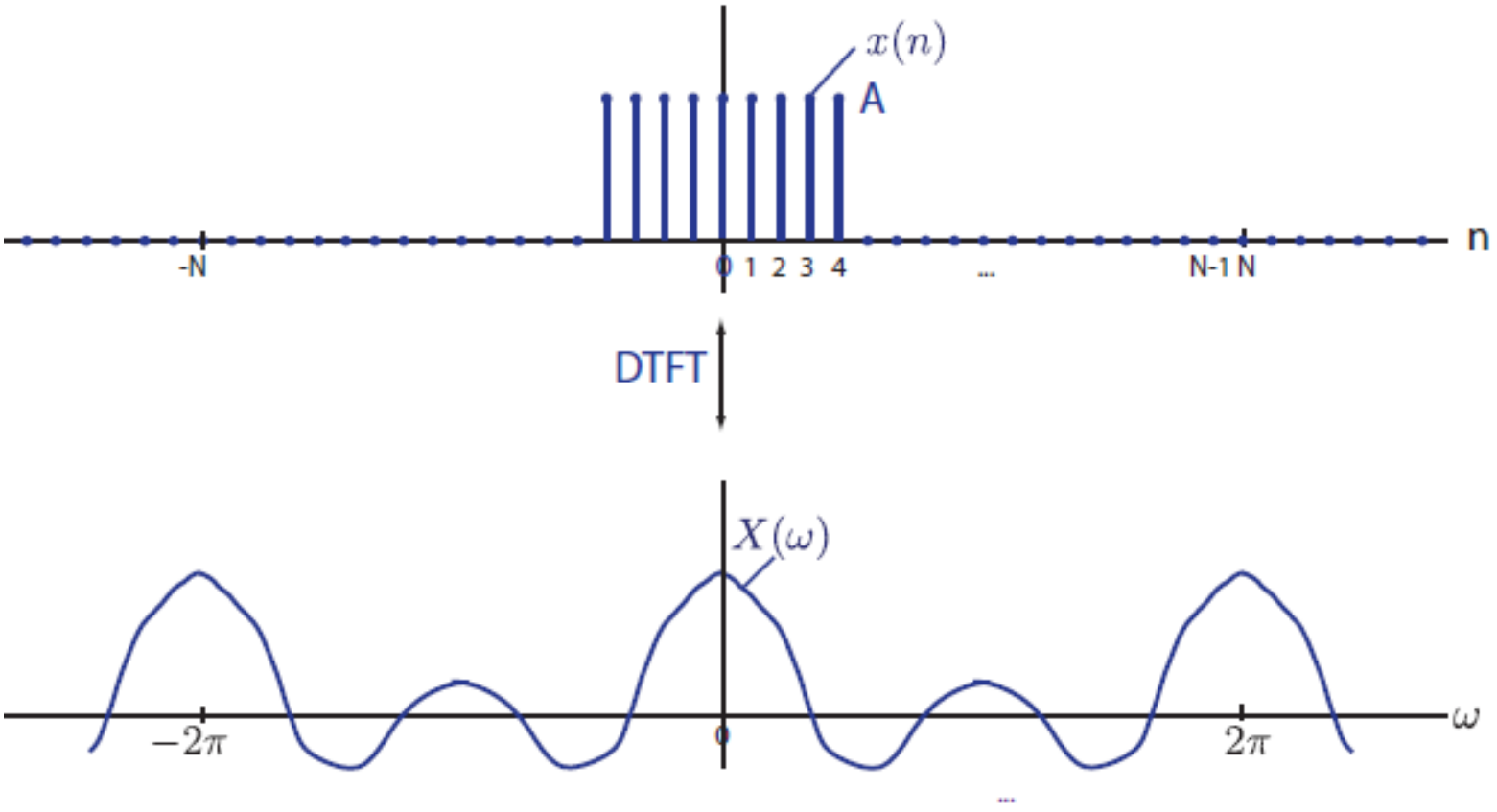
except we only take the coefficient values at $k = 0, 1, \ldots, N-1$.

Intuition

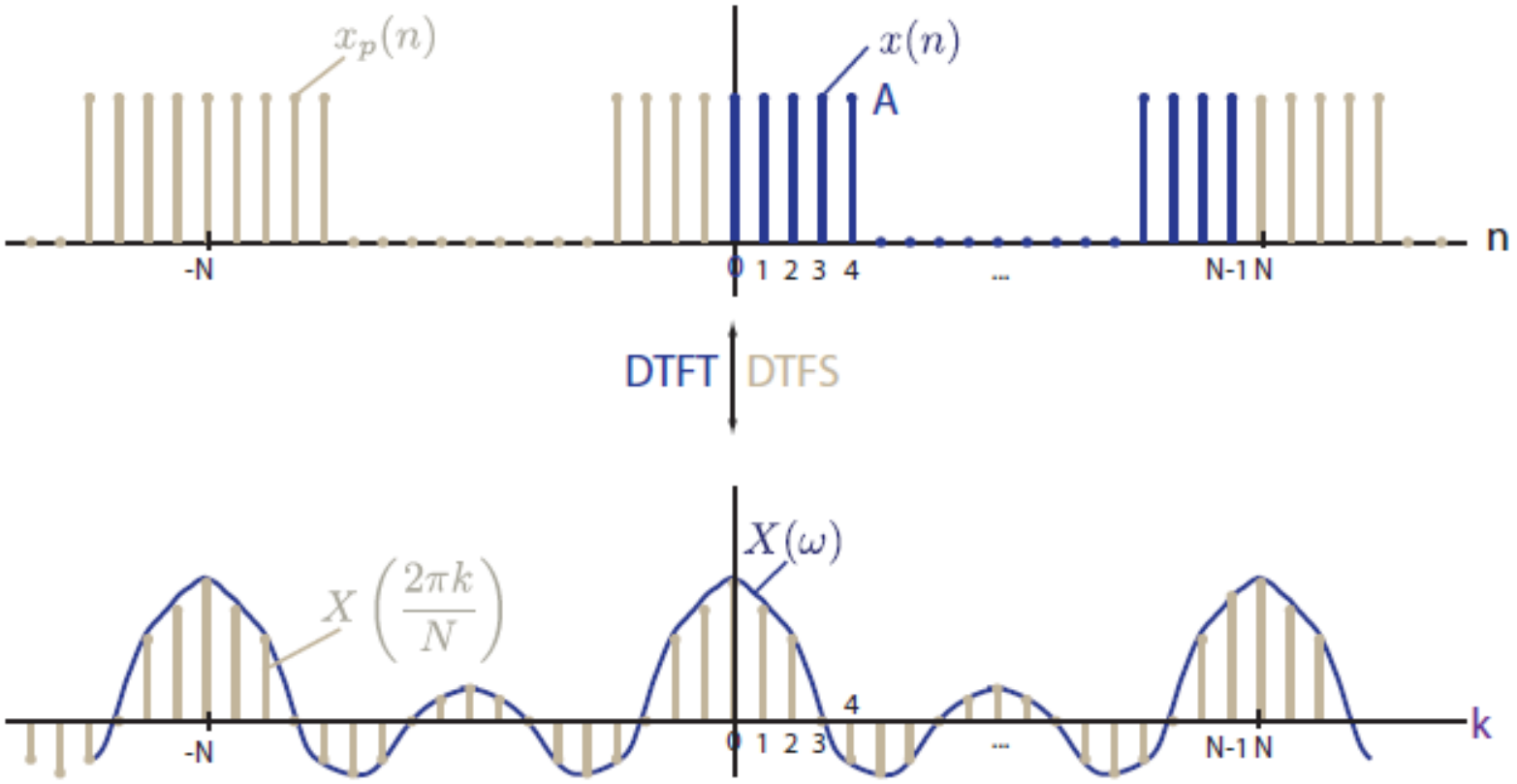
aperiodic + dst in time	DTFT $\longleftrightarrow$	cts + periodic in freq
$\downarrow$ periodic repetition		$\downarrow$ sampling
periodic + dst in time	DTFS $\longleftrightarrow$	dst + periodic in freq
one period of dst-time samples	DFT $\longleftrightarrow$	one period of dst-freq samples
$n = 0, 1, \ldots, N-1$		$k = 0, 1, \ldots, N-1$

- Therefore, we define the Discrete Fourier Transform (DFT) as being a computable transform that approximates the DTFT.

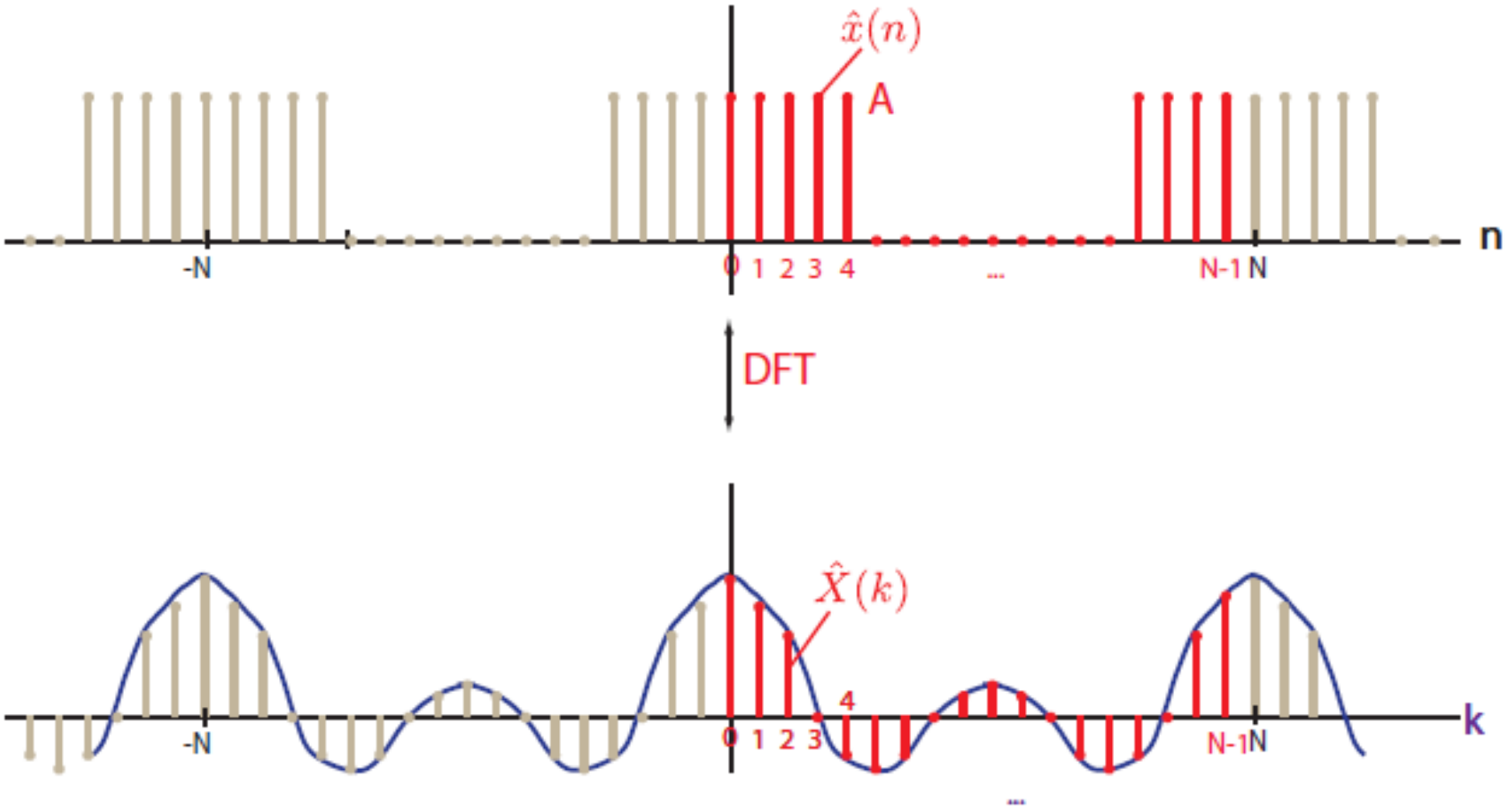
Intuition
Example



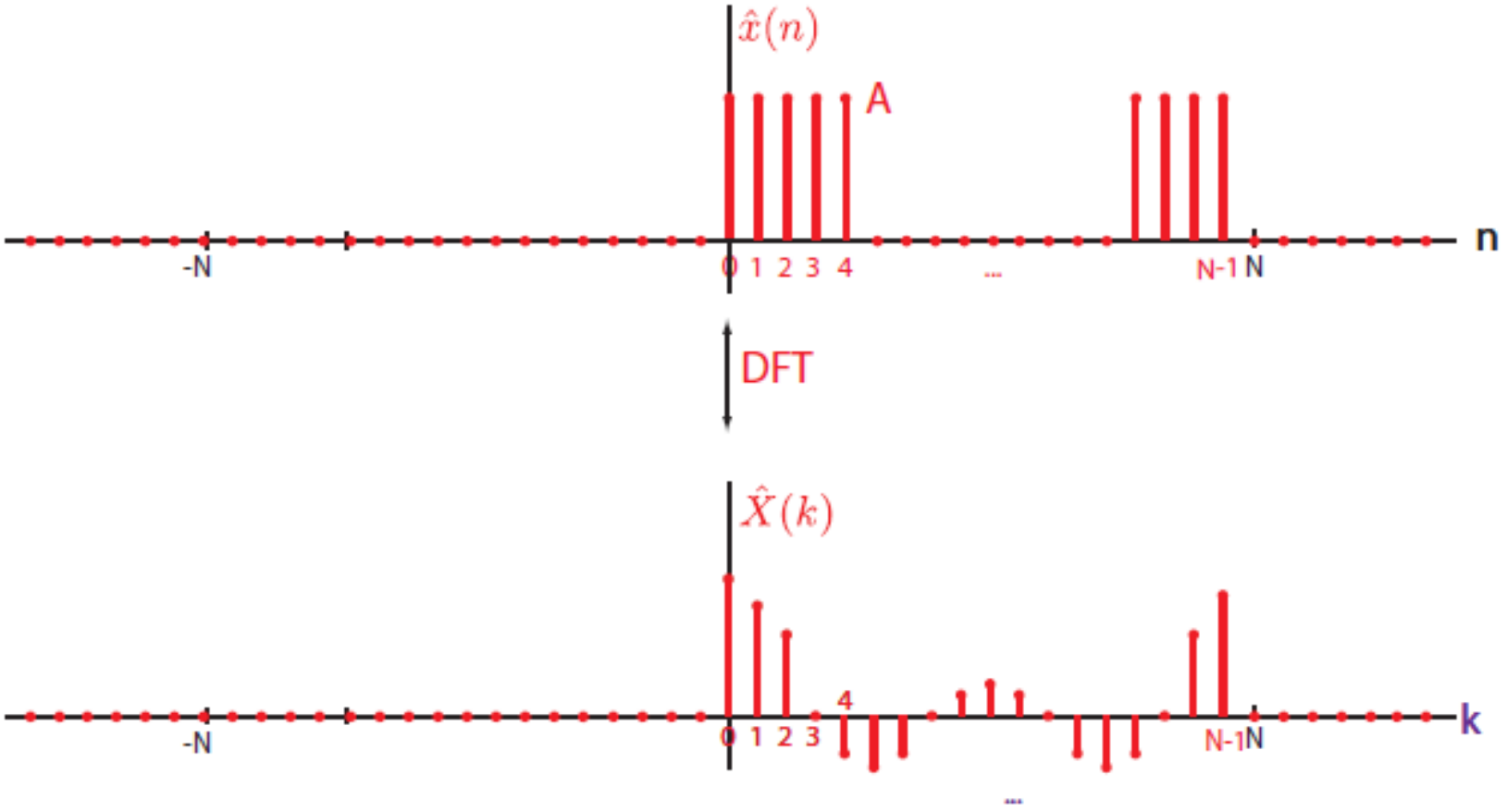
Intuition
Example



Intuition
Example



Intuition
Example



DTFT, DTFS and DFT

$$x(n) \text{ for all } n \quad \overset{\text{DTFT}}{\longleftrightarrow} \quad X(\omega) \text{ for all } \omega$$

$$\downarrow \text{ periodic repetition} \qquad \downarrow \text{ sampling}$$

$$x_p(n) = \sum_{l=-\infty}^{\infty} x(n+lN) \text{ for all } n \quad \overset{\text{DTFS}}{\longleftrightarrow} \quad X(k) = X(\omega)|_{\omega=\frac{2\pi}{N}k} \text{ for all } k$$

$$\hat{x}(n) \quad \overset{\text{DFT}}{\longleftrightarrow} \quad \hat{X}(k)$$

where

$$\hat{x}(n) = \begin{cases} x_p(n) & \text{for } n = 0, \ldots, N-1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$\hat{X}(k) = \begin{cases} X(k) & \text{for } k = 0, \ldots, N-1 \\ 0 & \text{otherwise} \end{cases}$$

Frequency Domain Sampling

- Recall, sampling in time results in a periodic repetition in frequency.

$$x(n) = x_a(t)|_{t=nT} \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad X(\omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_a(\omega + \frac{2\pi}{T}k)$$

- Similarly, sampling in frequency results in periodic repetition in time.

$$x_p(n) = \sum_{l=-\infty}^{\infty} x(n + lN) \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad X(k) = X(\omega)|_{\omega=\frac{2\pi}{N}k}$$

Frequency Domain Sampling and Reconstruction

- Therefore,

$$x(n) \overset{\mathcal{F}}{\longleftrightarrow} X(\omega)$$
$$x_p(n) \overset{\mathcal{F}}{\longleftrightarrow} X(k)$$

- Implications:
  - The samples of $X(\omega)$ can be used to reconstruct $x_p(n)$.

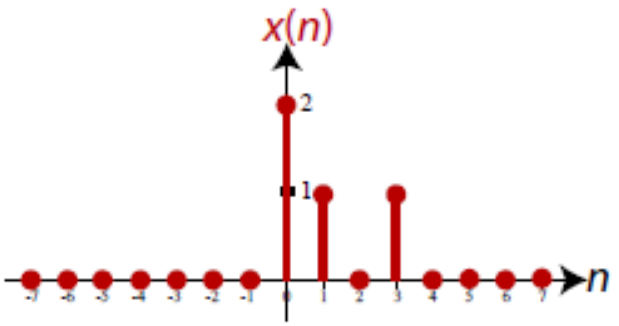
Frequency Domain Sampling and Reconstruction

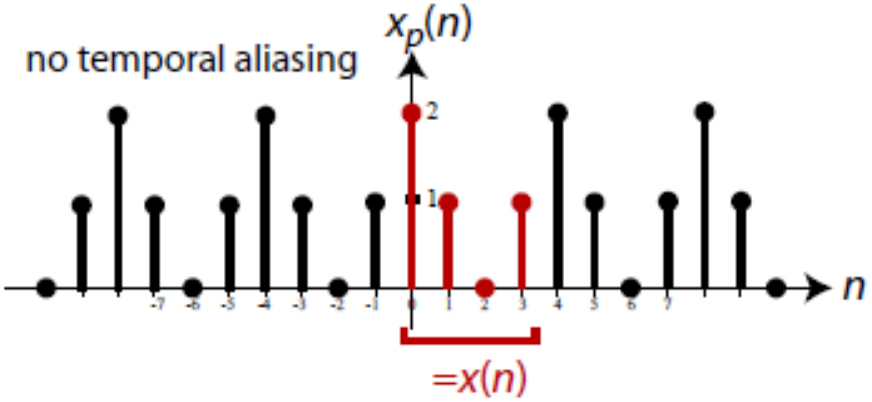
- Q: Can we reconstruct $x(n)$ from the samples of $X(\omega)$?
  - Can we reconstruct $x(n)$ from $x_p(n)$?
- A: Maybe.

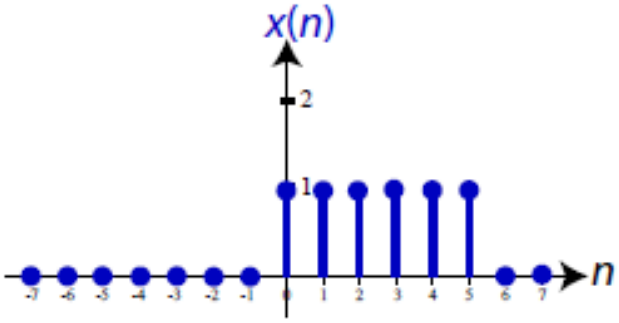
$$x_p(n) = \left[\sum_{l=-\infty}^{\infty} x(n + lN) \right]$$

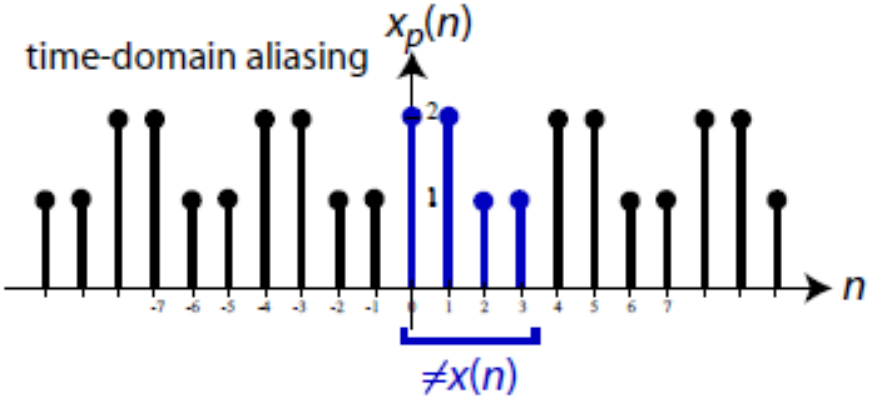
Frequency Domain Sampling and Reconstruction

$N = 4$



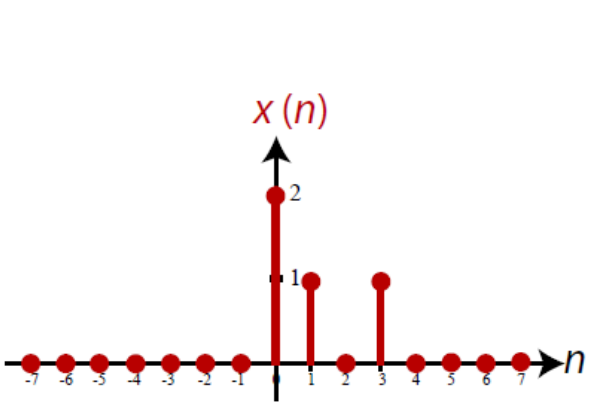


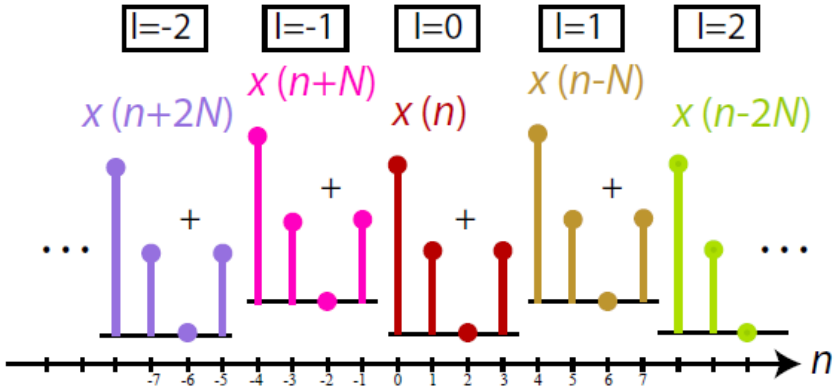




Frequency Domain Sampling and Reconstruction

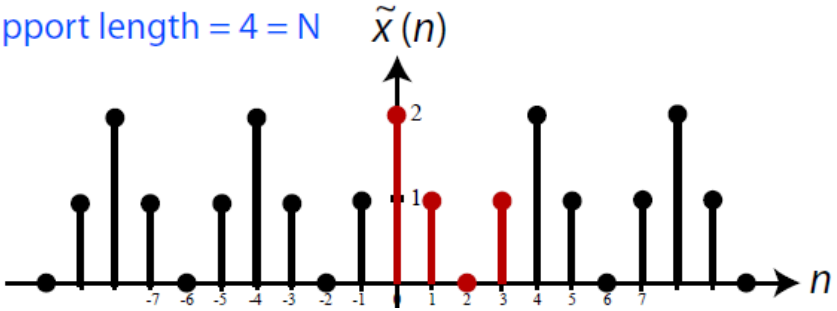
$N = 4$





no overlap

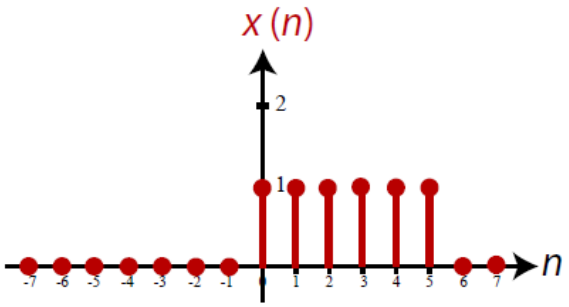
support length = 4 = N

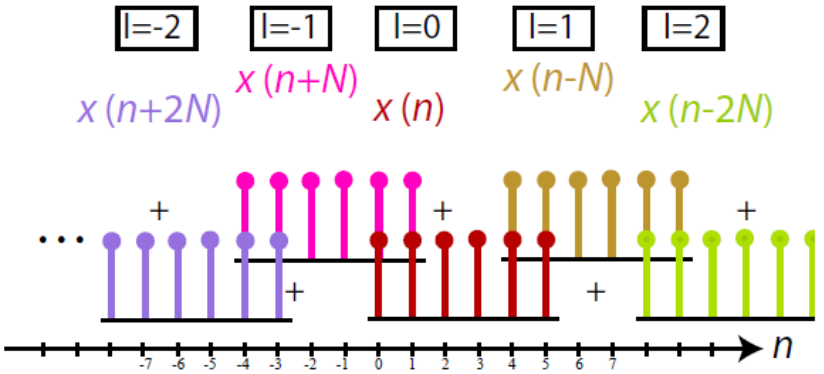


$= x\,(n)$

Frequency Domain Sampling and Reconstruction

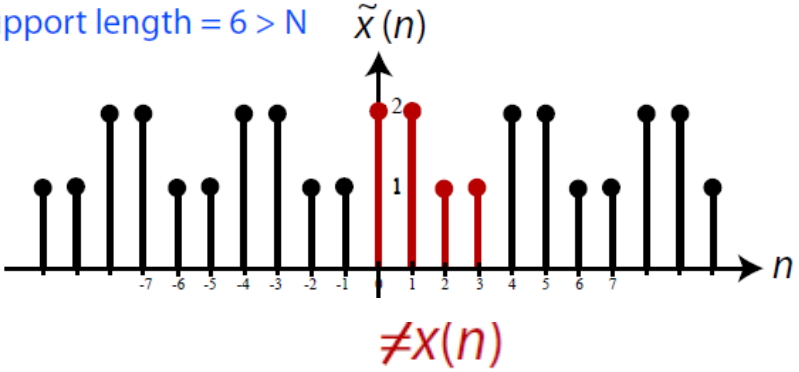
$N = 4$





overlap

support length = 6 > N



Frequency Domain Sampling and Reconstruction

- $x(n)$ can be recovered from $x_p(n)$ if there is no overlap when taking the periodic repetition.
- If x(n) is finite duration and non-zero in the interval $0 \leq n \leq L-1$, then

$$x(n) = x_p(n);\ 0 \leq n \leq N-1 \text{ when } N \geq L$$

- If $N < L$ then, $x(n)$ cannot be recovered from $x_p(n)$.
  - or equivalently $X(\omega)$ cannot be recovered from its samples $X(\frac{2\pi}{N}k)$ due to time-domain aliasing

The Discrete Fourier Transform Pair

- DFT and inverse-DFT (IDFT):

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi k \frac{n}{N}}, \quad k = 0, 1, \ldots, N-1$$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi k \frac{n}{N}}, \quad n = 0, 1, \ldots, N-1$$

- Note: we drop the $\hat{.}$ notation from now on.

DFT Example

Q: Determine the N-point DFT of the following sequence for $N \geq L$:

$$x(n) = \begin{cases} 1 & 0 \leq n \leq L-1 \\ 0 & otherwise \end{cases}$$

A: The DTFT of $x(n)$ is given by:

$$X(\omega) = \sum_{n=0}^{L-1} x(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{L-1} e^{-j\omega n} = \frac{\sin(\omega L/2)}{\sin(\omega/2)} e^{-j\omega(L-1)/2}$$

DFT Example

- Thus,

$$
\begin{aligned}
X(\omega) &= \frac{\sin(\omega L/2)}{\sin(\omega/2)} e^{-j\omega(L-1)/2} \\
X(k) &= \frac{\sin(\frac{2\pi k}{N} L/2)}{\sin(\frac{2\pi k}{N}/2)} e^{-j\frac{2\pi k}{N}(L-1)/2} \\
&= \frac{\sin(\pi kL/N)}{\sin(\pi k/N)} e^{-j\pi k(L-1)/N}
\end{aligned}
$$

DFT as linear transformation

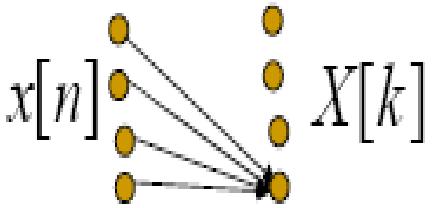
- Discrete Fourier transform (DFT) of a discrete-time signal $x[n]$ with finite extent $n \in [0, N\text{-}1]$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi nk}{N}} \qquad x[n] = \frac{1}{N} \sum_{n=0}^{N-1} X[k] e^{j\frac{2\pi nk}{N}}$$

- Twiddle factor:

$$W_N^k = e^{-j(2\pi/N)k} = \cos(2\pi k/N) + j\sin(2\pi k/N), \quad k = 0,1,\ldots,N-1$$

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \qquad k = 0,1,\ldots,N-1$$



DFT as linear transformation

- Let N-point vector x_N of the signal sequence x(n) , n = 0, 1, 2, ….(N-1) and N-point vector X_N of frequency samples and NxN matrix W_N as

$$\mathbf{x}_N = \begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix}, \qquad \mathbf{X}_N = \begin{bmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{bmatrix}$$

$$\mathbf{W}_N = \begin{bmatrix} 1 & 1 & 1 \cdots & 1 & \\ 1 & W_N & W_N^2 & \cdots & W_N^{N-1} \\ & W_N^2 & W_N^4 & \cdots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \vdots\vdots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \cdots & W_N^{(N-1)(N-1)} \end{bmatrix}$$

DFT as linear transformation

- With these definitions, the N-point DFT may be expressed in matrix form as, $\mathbf{X}_N = \mathbf{W}_N \mathbf{x}_N$
- The inverse DFT $\mathbf{x}_N = \mathbf{W}_N^{-1} \mathbf{X}_N$
- Or $\mathbf{x}_N = \frac{1}{N} \mathbf{W}_N^* \mathbf{X}_N$

$$\mathbf{W}_N^{-1} = \frac{1}{N} \mathbf{W}_N^* \qquad \mathbf{W}_N \mathbf{W}_N^* = N \mathbf{I}_N$$

I_N is an NxN identity matrix.

Example:

- Compute the DFT of the 4-point sequence

$$x(n) = (0 \quad 1 \quad 2 \quad 3)$$

$$\mathbf{W}_4 = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & W_4^1 & W_4^2 & W_4^3 \\ 1 & W_4^2 & W_4^0 & W_4^2 \\ 1 & W_4^3 & W_4^2 & W_4^1 \end{bmatrix} \qquad W_N = e^{-j2\pi/N}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \quad \text{then} \quad \mathbf{X}_4 = \mathbf{W}_4\mathbf{x}_4 = \begin{bmatrix} 6 \\ -2+2j \\ -2 \\ -2-2j \end{bmatrix}$$

Example

Determine Discrete Fourier Transform (DFT) of the following sequence:

$x[n] = a^n U[n]$ $\qquad |a| < 1$

$$x[k] = \sum_{n=-\infty}^{\infty} x(nT) e^{-j2k\pi \frac{n}{N}} = \sum_{n=-\infty}^{\infty} a^n U_s(n) e^{-j2k\pi \frac{n}{N}}$$

$$x[k] = \sum_{n=0}^{\infty} a^n e^{-j2k\pi \frac{n}{N}} = \sum_{n=0}^{\infty} (ae^{-j\frac{2k\pi}{N}})^n$$

$$x[k] = \frac{1}{1 - ae^{-j\frac{2k\pi}{N}}}$$

k = 0 , 1 , 2 , 3 , ……., N-1

Example

Determine Discrete Fourier Transform (DFT) of the following sequence:

x[n] = $a^n U[n+3]$ $\qquad |a| < 1$

$$x[k] = \sum_{n=-\infty}^{\infty} x(nT) e^{-j2k\pi \frac{n}{N}} = \sum_{n=-\infty}^{\infty} a^n U_s(n+3) e^{-j2k\pi \frac{n}{N}}$$

$$x[k] = \sum_{n=-3}^{\infty} a^n e^{-j2k\pi \frac{n}{N}} = \sum_{n=0}^{\infty} (ae^{-j\frac{2k\pi}{N}})^n + \sum_{n=-3}^{-1} (ae^{-j\frac{2k\pi}{N}})^n$$

$$x[k] = \frac{a^{-3} e^{j6k\pi}}{1 - ae^{-j\frac{2k\pi}{N}}}$$

k = 0 , 1 , 2 , 3 , ……, N-1

DFT Properties

- Periodicity:
  - If x(n) and X(k) are N-point DFT pairs

$$x(n+N) = x(n) \quad \text{for all } n$$

$$X(k+N) = X(k) \quad \text{for all } k$$

- Linearity:

$$ax_1[n] + bx_2[n] \overset{DFT}{\leftrightarrow} aX_1[k] + bX_2[k]$$

DFT Properties

- The properties of the DFT are different from those typical of the DTFS and DTFT because they are circular in nature.

- That is, they apply to the periodic repetition of the signal.

Important DFT Properties

Property	Time Domain	Frequency Domain
Notation:	$x(n)$	$X(k)$
Periodicity:	$x(n) = x(n+N)$	$X(k) = X(k+N)$
Linearity:	$a_1 x_1(n) + a_2 x_2(n)$	$a_1 X_1(k) + a_2 X_2(k)$
Time reversal	$x(N-n)$	$X(N-k)$
Circular time shift:	$x((n-l))_N$	$X(k)e^{-j2\pi kl/N}$
Circular frequency shift:	$x(n)e^{j2\pi ln/N}$	$X((k-l))_N$
Complex conjugate:	$x^*(n)$	$X^*(N-k)$
Circular convolution:	$x_1(n) \otimes x_2(n)$	$X_1(k)X_2(k)$
Multiplication:	$x_1(n)x_2(n)$	$\frac{1}{N} X_1(k) \otimes X_2(k)$
Parseval's theorem:	$\sum_{n=0}^{N-1} x(n)y^*(n)$	$\frac{1}{N} \sum_{k=0}^{N-1} X(k)Y^*(k)$

Circular Properties

- Circular operations: apply the transformation on the periodic repetition of x(n) and then obtain the final result by taking points for $n = 0, 1, \dots, N-1$

- Often use the modulo notation:

$$(n)_N = n \mod N = \text{remainder of } n/N$$

Modulo Indices and Periodic Repetition

$$(n)_N = n \mod N = \text{remainder of } n/N$$

Example: $N = 4$

n	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8
$(n)_4$	0	1	2	3	0	1	2	3	0	1	2	3	0

$$\frac{n}{N} = \text{integer} + \frac{\boxed{\text{nonneg integer} < N}}{N}$$

$$\frac{5}{4} = 1 + \frac{1}{4} \qquad \frac{-2}{4} = -1 + \frac{2}{4}$$

Modulo Indices and Periodic Repetition

$$(n)_N = n \mod N = \text{remainder of } n/N$$

Example: $N = 4$

n	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8
$(n)_4$	0	1	2	3	0	1	2	3	0	1	2	3	0

$x((n))_4$ will be periodic with period 4. The repeated pattern will be consist of: $\{x(0), x(1), x(2), x(3)\}$.

Thus, $x((n))_N$ is a periodic signal comprised of the following repeating pattern: $\{x(0), x(1), \cdots x(N-2), x(N-1)\}$.

Circular Symmetry and Convolution

- Circular Symmetry:
  - circular time reversal: $x((-n))_N = x(N-n)$
  - circularly even: $x(N-n) = x(n)$
  - circularly odd: $x(N-n) = -x(n)$

- Circular Convolution:

$$x_3(m) = \sum_{n=0}^{N-1} x_1(n) x_2((m-n))_N, \quad m = 0, 1, \ldots, N-1$$

DFT Properties

Circular Convolution:

$$x_1(n) \overset{\text{DFT}}{\underset{N}{\longleftrightarrow}} X_1(k)$$

$$x_2(n) \overset{\text{DFT}}{\underset{N}{\longleftrightarrow}} X_2(k)$$

$$x_1(n) \textcircled{N} x_2(n) \overset{\text{DFT}}{\underset{N}{\longleftrightarrow}} X_1(k)X_2(k)$$

DFT Properties

Multiplication of 2 sequence:

Suppose we have 2 finite-duration sequences of length N, $x_1(n)$ and $x_2(n)$

$$X_3(k) = X_1(k)X_2(k) \qquad k = 0, 1, \ldots, N-1$$

Convolution Sum

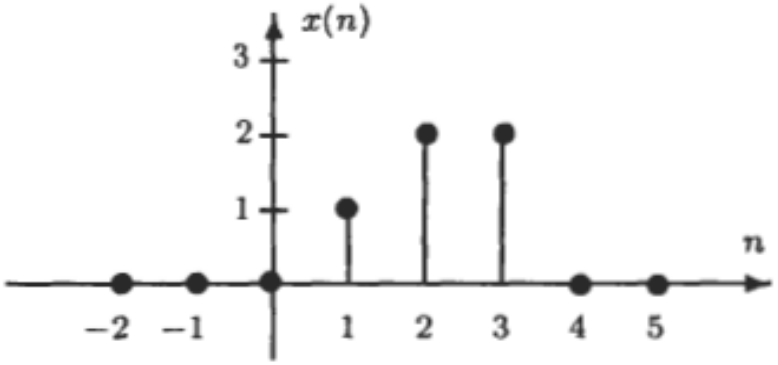
$$x_3(m) = \sum_{n=0}^{N-1} x_1(n)x_2((m-n))_N \qquad m = 0, 1, \ldots, N-1$$

called Circular Convolution

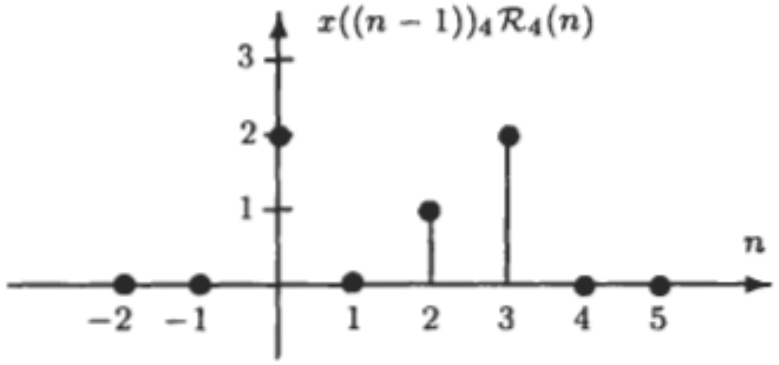
$$\tilde{x}(n) = x(n \bmod N) \equiv x((n))_N \qquad (n \bmod N) = ((n))_N = l$$

For example, $((13))_8 = 5$ and $((-6))_8 = 2$.

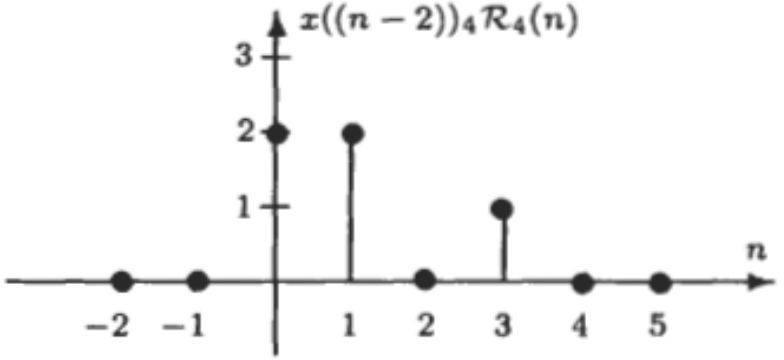
DFT Properties



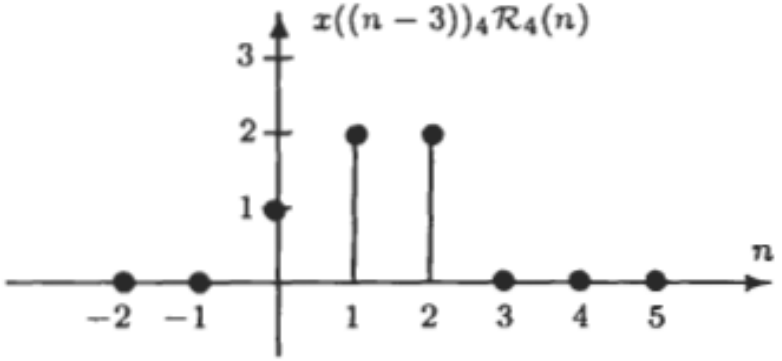
(*a*) A discrete-time signal of length $N = 4$.



(*b*) Circular shift by one.



(*c*) Circular shift by two.



(*d*) Circular shift by three.

DFT Properties

$$x(n-k, \text{modulo } N) \equiv x((n-k))_N$$

For example, if $k = 2$ and $N = 4$, we have

$$x'(n) = x((n-2))_4$$

which implies that

$$x'(0) = x((-2))_4 = x(2)$$

$$x'(1) = x((-1))_4 = x(3)$$

$$x'(2) = x((0))_4 = x(0)$$

$$x'(3) = x((1))_4 = x(1)$$

DFT Properties

Example:

Perform the circular convolution of the following 2 sequence:

$$x_1(n) = \{2, 1, 2, 1\} \quad (\uparrow \text{ at } n=0) \qquad x_2(n) = \{1, 2, 3, 4\} \quad (\uparrow \text{ at } n=0)$$

m=0

$$x_3(0) = \sum_{n=0}^{3} x_1(n)x_2((-n))_N$$

= 14

m=1

$$x_3(1) = \sum_{n=0}^{3} x_1(n)x_2((1-n))_4$$

= 16

m=2

$$x_3(2) = \sum_{n=0}^{3} x_1(n)x_2((2-n))_4$$

= 14

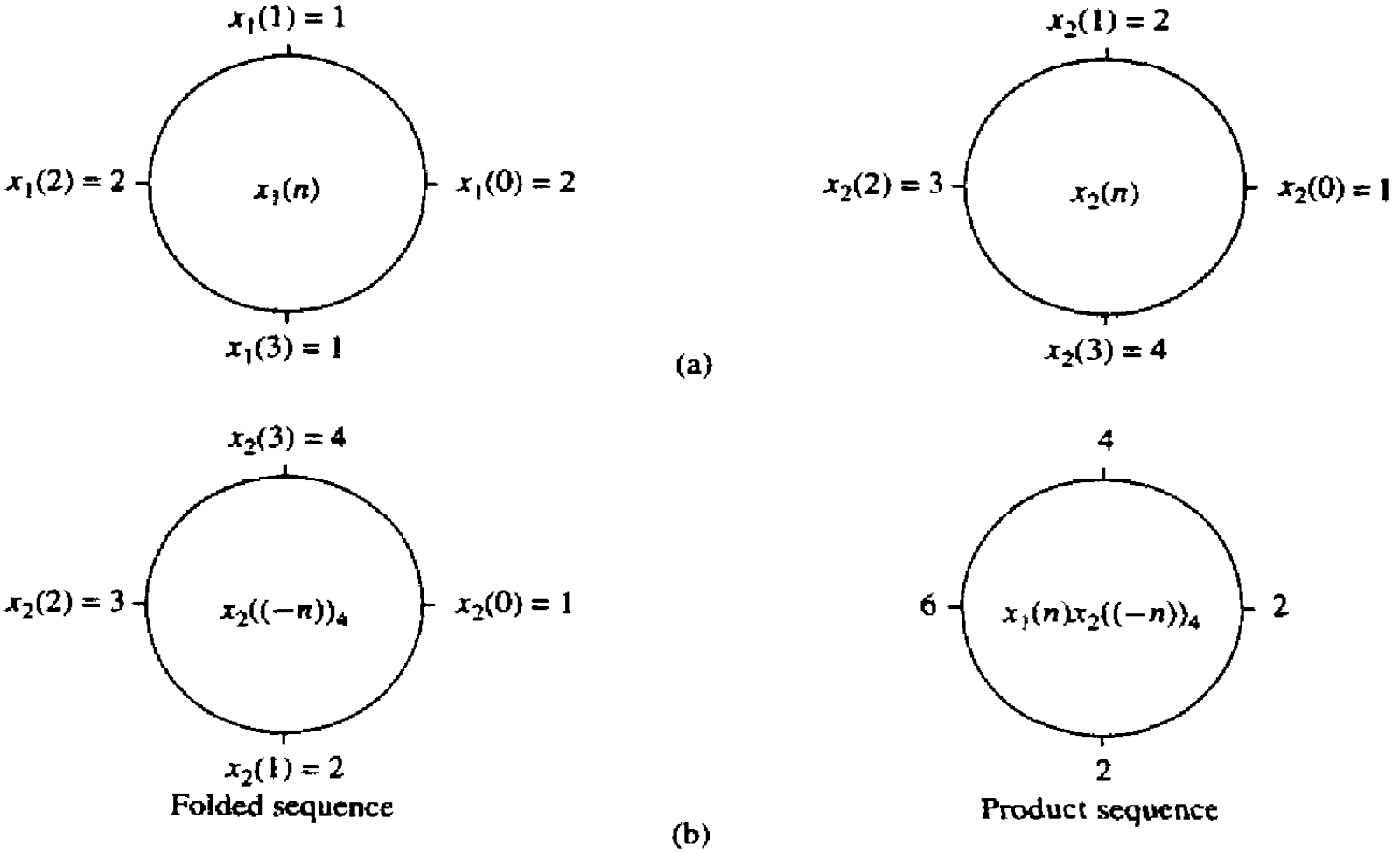
m=3

$$x_3(3) = \sum_{n=0}^{3} x_1(n)x_2((3-n))_4$$

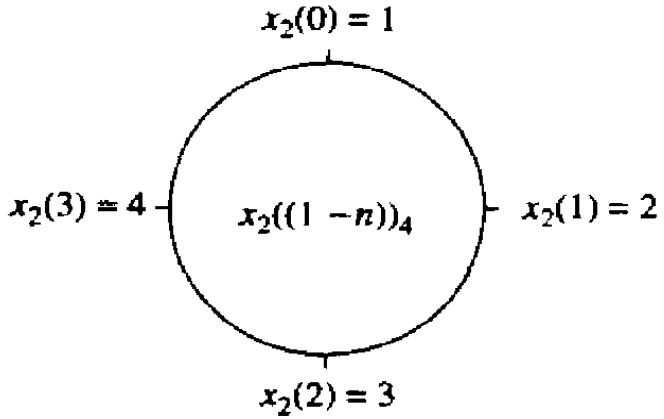
= 16

$$x_3(n) = \{14, 16, 14, 16\} \quad (\uparrow \text{ at } n=0)$$

DFT Properties

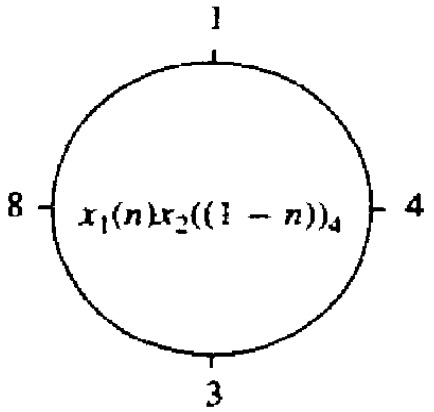


DFT Properties

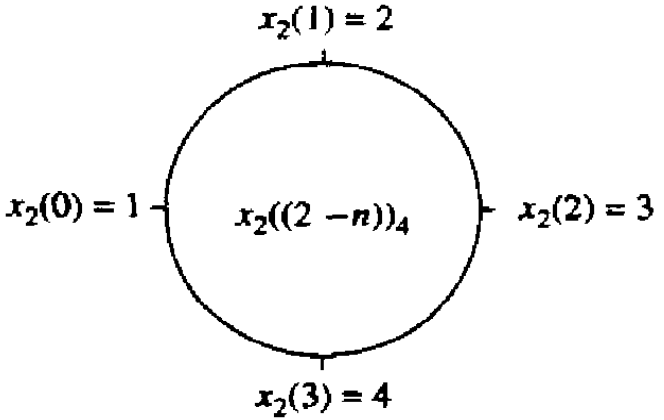


Folded sequence rotated by one unit in time

(c)

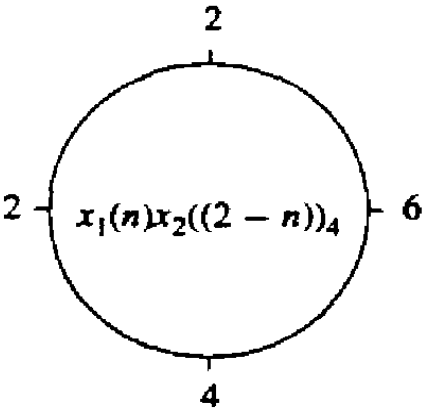


Product sequence



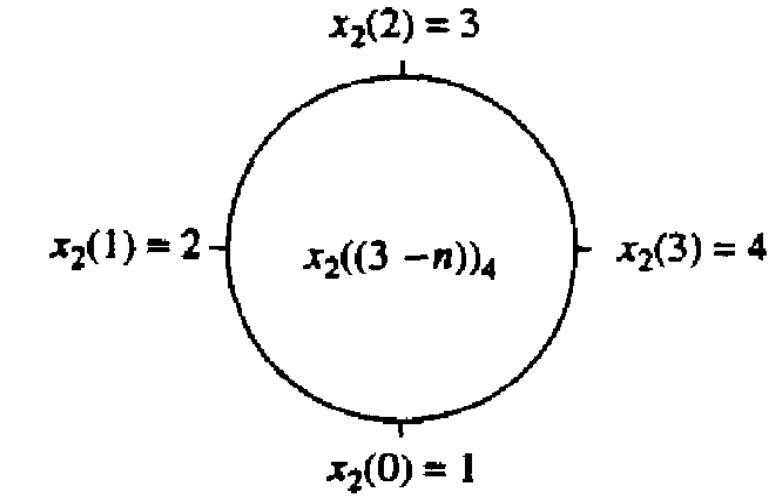
Folded sequence rotated by two units in time

(d)



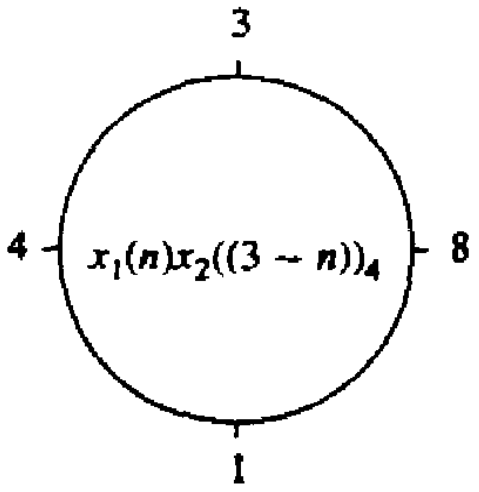
Product sequence

DFT Properties



Folded sequence rotated by three units in time

(e)



Product sequence

Circular Convolution

DFT Properties

Example:

Using DFT and IDFT determine the sequence $x_3(n)$ corresponding to the circular convolution of the sequences $x_1(n)$ and $x_2(n)$ given in the previous example.

$$X_1(k) = \sum_{n=0}^{3} x_1(n) e^{-j2\pi nk/4} \qquad k = 0, 1, 2, 3$$

$$= 2 + e^{-j\pi k/2} + 2e^{-j\pi k} + e^{-j3\pi k/2}$$

$$X_1(0) = 6 \qquad X_1(1) = 0 \qquad X_1(2) = 2 \qquad X_1(3) = 0$$

$$X_2(k) = \sum_{n=0}^{3} x_2(n) e^{-j2\pi nk/4} \qquad k = 0, 1, 2, 3$$

$$= 1 + 2e^{-j\pi k/2} + 3e^{-j\pi k} + 4e^{-j3\pi k/2}$$

$$X_2(0) = 10 \qquad X_2(1) = -2 + j2 \qquad X_2(2) = -2 \qquad X_2(3) = -2 - j2$$

DFT Properties

Example: (cont.)

$$X_3(k) = X_1(k)X_2(k)$$

$X_3(0) = 60$	$X_3(1) = 0$	$X_3(2) = -4$	$X_3(3) = 0$

$$x[n] = \frac{1}{N}\sum_{n=0}^{N-1} X[k]e^{j\frac{2\pi nk}{N}}$$

IDFT ⇒

$$x_3(n) = 1/4 \sum_{k=0}^{3} X_3(k)e^{j2\pi nk/4} \qquad n = 0, 1, 2, 3$$

$$= \tfrac{1}{4}(60 - 4e^{j\pi n})$$

$x_3(0) = 14$	$x_3(1) = 16$	$x_3(2) = 14$	$x_3(3) = 16$

DFT Properties

Find the 10-point inverse DFT of

$$X(k) = \begin{cases} 3 & k = 0 \\ 1 & 1 \le k \le 9 \end{cases}$$

To find the inverse DFT, note that $X(k)$ may be expressed as follows:

$$X(k) = 1 + 2\delta(k) \qquad 0 \le k \le 9$$

Written in this way, the inverse DFT may be easily determined. Specifically, note that the inverse DFT of a constant is a unit sample:

$$x_1(n) = \delta(n) \overset{DFT}{\Longleftrightarrow} X_1(k) = 1$$

Similarly, the DFT of a constant is a unit sample:

$$x_2(n) = 1 \overset{DFT}{\Longleftrightarrow} X_2(k) = N\delta(k)$$

Therefore, it follows that

$$x(n) = \tfrac{1}{5} + \delta(n)$$